

Manufacturing Energy & Carbon Analysis

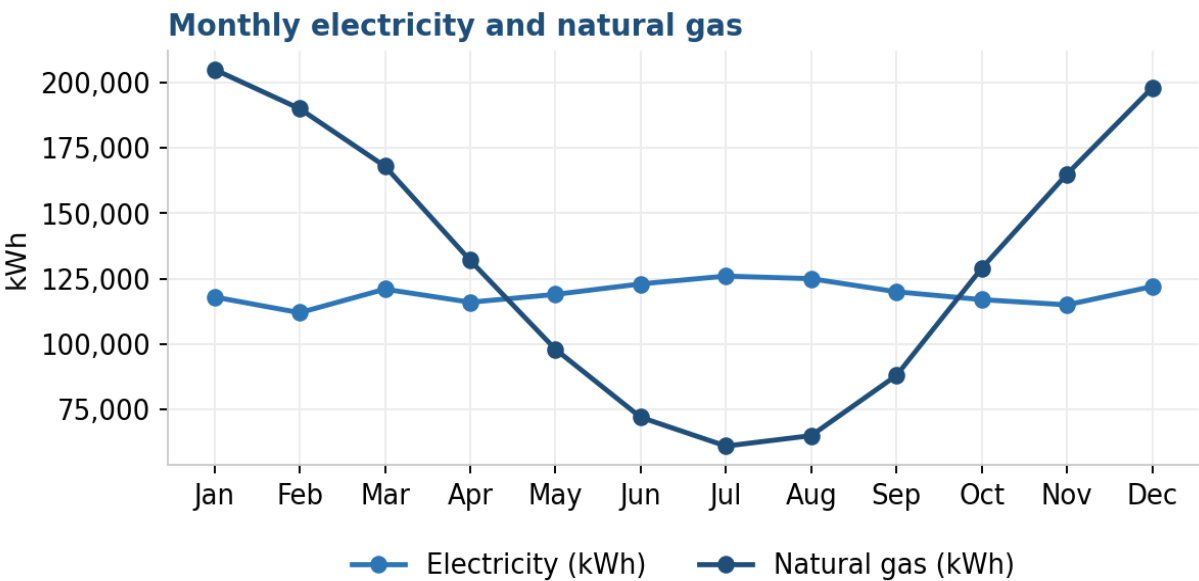
Executive summary

This synthetic case demonstrates the workflow from raw operational data to a management-ready analysis. All figures are illustrative and not real client data.

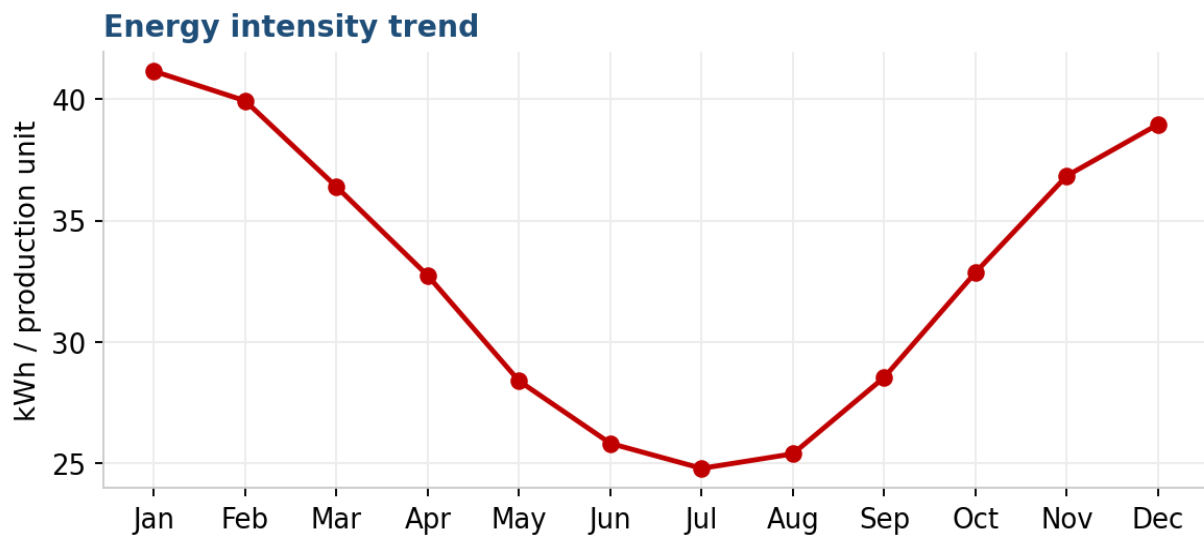
KPI	Illustrative result	Unit
Total energy	3,970,300	kWh
Total emissions (Scope 1 + 2)	1,083.2	tCO2e
Energy intensity	32.46	kWh/unit
Carbon intensity	8.86	kgCO2e/unit

Energy & activity analysis

Electricity is relatively stable through the year (112,000–126,000 kWh/month), while natural gas swings 3.4x between its July trough (61,000 kWh) and January peak (205,000 kWh). This shows a clear heating/process-driven seasonal signature.

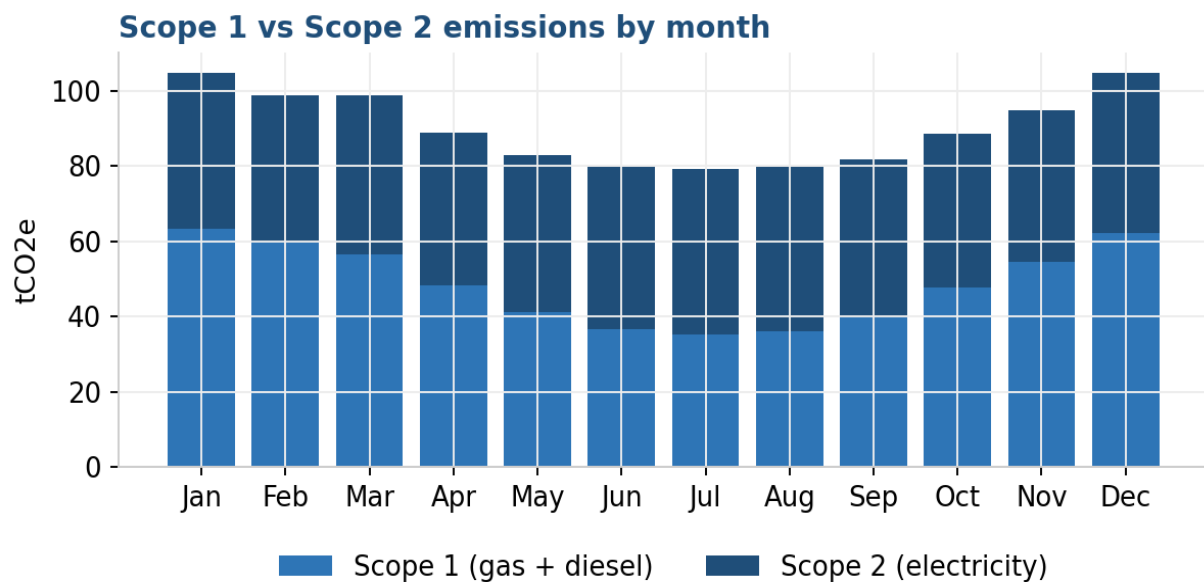


Production-normalised energy intensity ranges from 24.8 kWh/unit in July to 41.2 kWh/unit in January.



Carbon / KPI analysis

Scope 1 is screened from gas and diesel; Scope 2 from purchased electricity, using illustrative factors. The workbook separates activity data and factors to keep assumptions transparent and auditable.



Key insight: production output and total energy show a weak, negative correlation ($r = -0.70$) across the year. That means the seasonal swing is driven by fixed/weather-related loads (heating, baseload), not by how much is produced.

Data quality

Real work would reconcile bills/meter exports, define boundaries, document exclusions and agree factors before reporting. In this workbook, the Clean Data tab is linked live to Raw Data (not retyped) and carries an automated flag column that screens for missing values and statistical outliers.

Recommendations

1. Initiate a sub-meter process, building/HVAC and fleet loads separately. This is the single change most likely to explain the weak production–energy correlation and unlock real savings targeting.
2. Review electricity baseload and operating schedules; electricity barely moves with output, suggesting always-on equipment or scheduling inefficiency worth a walk-through.
3. Investigate the natural-gas seasonal swing against heating-degree-days to separate weather effects from process/equipment inefficiency.

Scope note: This portfolio demonstrates data preparation, screening calculations, KPI development and reporting. It is not independent GHG verification or assurance, a statutory energy audit, legal advice, or a guarantee of savings. All figures in this document are illustrative, synthetic data for demonstration purposes only.